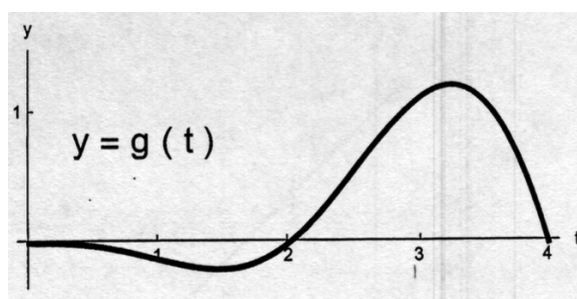


Problem 2

Problem 2

Problem 1 The graph of g , a continuous function on $[0, 4]$, is shown in the figure. Let $A(x) = \int_0^x g(t) dt$, for $0 \leq x \leq 4$.



(a) Circle the correct statement about $A(2)$.

- (i) $A(2) = 0$;
- (ii) $A(2) > 0$;
- (iii) $A(2) < 0$;
- (iv) none of the previous answers.

(b) Circle the correct statement about $A(3.8)$.

- (i) $A(3.8) = 0$;
- (ii) $A(3.8) > 0$;
- (iii) $A(3.8) < 0$;
- (iv) none of the previous answers.

(c) Circle the correct statement about $A'(3.8)$.

- (i) $A'(3.8) = 0$;
- (ii) $A'(3.8) > 0$;
- (iii) $A'(3.8) < 0$;

Problem 2

(iv) *none of the previous answers.*

(d) Find the solution of the following initial value problem: $y'(x) = g(x)$ and $y(0) = 2$.

(i) $y(x) = g(x)$;

(ii) $y(x) = g(x) + 2$;

(iii) $y(x) = A(x)$;

(iv) $y(x) = A(x) + 2$;

(v) $y(x) = g'(x)$;

(vi) $y(x) = g'(x) + 2$;

(vii) *none of the previous answers.*

[2]

(e) Find the expression for $\int_0^4 |g(t)| dt$.

(i) $A(4)$;

(ii) $A(2) - A(4)$;

(iii) $A(4) - A(2)$;

(iv) $A(4) - 2A(2)$;

(v) *none of the previous answers.*

(f) Find the midpoint Riemann sum for the function g on the interval $[0, 4]$ with $n = 2$ (the number of subintervals).

(i) $g(1) + g(3)$;

(ii) $g(0) + g(2)$;

(iii) $g(2) + g(4)$;

(iv) $2g(1) + 2g(3)$;

(v) $2g(0) + 2g(2)$;

(vi) $2g(2) + 2g(4)$;

(vii) *none of the previous answers.*